

Abdullah Syed

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EDUCATION:

Toronto Metropolitan University

Toronto, Canada

Electrical Engineering – Bachelor of Engineering

2019-2024

Relevant Courses: • Microelectronic Circuits • Digital Signal Processing • Digital Control Systems • Communication Systems • Energy Conversion • Machine Learning • Object Oriented Design in Java • Digital Systems • Robotics

SKILLS & INTERESTS:

- **Languages Spoken:** English, Urdu, Hindi
- **Programming Languages:** Java, C, MATLAB, JavaScript, HTML, CSS, JSX, Python
- **Technologies:** Multisim, Intel Quartus, Simulink, Google Collab, ReactJS, Node.js, Microsoft Office, Fusion 360

PROJECTS:

Non Intrusive Load Monitoring (NILM) Smart Energy Meter - September 2023 - April 2024

[Project Link](#)

- Developed a smart energy meter using ESP32 microcontroller, enabling real-time simultaneous monitoring of multiple household appliances without requiring intrusive hardware modifications.
- Implemented a NILM (Non-Intrusive Load Monitoring) algorithm to analyze energy consumption patterns and detect which appliances were active at specific times, providing insights into usage patterns.
- Leveraged Firebase Realtime Database to store and retrieve appliance data, ensuring seamless synchronization and accessibility across devices. Optimized functionality by deploying Firebase Cloud Functions, enabling serverless computing to automate tasks and improve overall system performance.
- Designed and optimized a custom PCB using AutoCAD software Fusion 360 for the energy meter circuit, ensuring efficient layout and performance.

Design of PID Controller for Servo Positioning Module - March 2023

[Project Link](#)

- Implemented a PID controller for a simulated and real-time DC servo motor to enhance the closed-loop response of the system to a square wave input, demonstrating expertise in control system design and optimization.
- Analyzed response specifications for both uncompensated and compensated systems, comparing rise time, settling time, maximum overshoot, and steady-state error to assess the effectiveness of the control strategies.

Linear Voltage-Controlled Frequency Function Generator - October 2022

[Project Link](#)

- Designed and crafted a device capable of producing symmetrical triangular and square waves by employing a systematic approach that encompassed stages of circuit design analysis, Multisim simulation, and physical implementation on a breadboard.
- Incorporated two user-selectable frequency ranges: Range 1 (100 Hz - 4200 Hz) and Range 2 (20 Hz - 840 Hz), as well as user-controlled gain ranging from 0-4Vp providing flexibility in wave generation.
- Utilized unit testing methodologies to identify, troubleshoot and resolve issues within the circuit design and implementation. Employed bypass capacitor circuits to minimize noise and ensure stable performance.

PROFESSIONAL EXPERIENCE:

Sheen For She Foundation - [Website](#)

Toronto, Canada

Event Coordinator and Volunteer Leader

Mar 2020 – Present

- Actively contributed to the marketing efforts of Sheen for She Foundation, raising awareness about its services and initiatives.
- Played a vital role in planning and executing successful events, ensuring smooth logistics and effective implementation.
- Developed strong leadership skills by effectively managing and coordinating volunteers and engaging with clients in a compassionate and empathetic manner.
- Effectively communicated the mission and impact of Sheen for She Foundation to stakeholders and the community, fostering a greater understanding and support for the organization's goals.

Toronto Metropolitan University Recreation - [Website](#)

Toronto, Canada

Programming Strategist & Supervisor

Sept 2023 – April 2024

- Utilized strategic analysis and metric tracking to optimize the Drop-In Sports program, enhancing student participation rates and benefits.
- As a committed supervisor, ensured the safety and inclusivity of daily drop-in sessions and special events, taking the lead in promptly addressing injuries and managing various emergent situations.
- Acted as a university ambassador, delivering exceptional support to clients during special events.